

MTH313M Quiz Part A (April 7, 2026)

6:45 pm - 7:30 pm (+15 minutes for DAP students)

Instructions:

1. Write your answers in the designated place. If your answer is not legible, you shall not get credit.
2. DO NOT attach any additional sheet to this page.
3. Read the question stem and answer the sub-questions separately in the space provided.
4. Provide all details/justification including calculations.

Name:

Roll No.:

Question 1. (4 marks) Let $X_t = \sum_{j=1}^{N_t} Y_j, t \geq 0$ be a Compound Poisson Process with rate $\lambda > 0$ and component distribution F (i.e. $Y_1, Y_2, \dots$ are i.i.d. with common DF F). Compute $Cov(X_s, X_t)$ for $0 \leq s \leq t < \infty$.

Question 2. (6 marks) Suppose that shocks occur in a system according to a Poisson Process with rate $\lambda > 0$. Further assume that the shocks independently causes the system to fail with probability $p \in (0, 1)$ and that the system continues to work till it fails. Let M denote the number of shocks finally causing the system to fail and T denote the time of failure. Compute $\mathbb{P}(M = m | T = t)$ for $m = 1, 2, \dots, t \in (0, \infty)$.